Planets and Pluto: Physical Characteristics

This table contains selected physical characteristics of the planets and Pluto.

Planet	Equatorial Radius	Mean Radius	Mass	Bulk Density	Sidereal Rotation Period	Sidereal Orbit Period	V(1,0)	Geometric Albedo	Equatorial Gravity	Escape Velocity
	(km)	(km)	(x 10^{24} kg)	(g cm^{-3})	(d)	(y)	(mag)		(m s^{-2})	(km s^{-1})
Mercury	2440.53 [D] ±0.04	2439.4 [D] ±0.1	0.330114 [E] ±0.000021	5.4291 [*] ±0.0007	58.6462 [C]	0.2408467 [B]	-0.60 [E] ±0.10	0.106 [B]	3.70 [*]	4.25 [*]
Venus	6051.8 [D] ±1.0	6051.8 [D] ±1.0	4.86747 [G] ±0.00023	5.243 [*] ±0.003	-243.018 [C]	0.61519726 [B]	-4.47 [E] ±0.07	0.65 [B]	8.87 [*]	10.36 [*]
Earth	6378.1366 [D] ±0.0001	6371.0084 [D] ±0.0001	5.97237 [H] ±0.00028	5.5136 [*] ±0.0003	0.99726968 [B]	1.0000174 [B]	-3.86 [B]	0.367 [B]	9.80 [*]	11.19 [*]
Mars	3396.19 [D] ±0.1	3389.50 [D] ±0.2	0.641712 [I] ±0.000030	3.9341 [*] ±0.0007	1.02595676 [C]	1.8808476 [B]	-1.52 [B]	0.150 [B]	3.71 [*]	5.03 [*]
Jupiter	71492 [D] ±4	69911 [D] ±6	1898.187 [J] ±0.088	1.3262 [*] ±0.0003	0.41354 [C]	11.862615 [B]	-9.40 [B]	0.52 [B]	24.79 [*]	60.20 [*]
Saturn	60268 [D] ±4	58232 [D] ±6	568.336 [K] ±0.026	0.6871 [*] ±0.0002	0.44401 [C]	29.447498 [B]	-8.88 [B]	0.47 [B]	10.44 [*]	36.09 [*]
Uranus	25559 [D] ±4	25362 [D] ±7	86.8127 [L] ±0.0040	1.270 [*] ±0.001	-0.71833 [C]	84.016846 [B]	-7.19 [B]	0.51 [B]	8.87 [*]	21.38 [*]
Neptune	24764 [D] ±15	24622 [D] ±19	102.4126 [M] ±0.0048	1.638 [*] ±0.004	0.67125 [C]	164.79132 [B]	-6.87 [B]	0.41 [B]	11.15 [*]	23.56 [*]
Pluto	1188.3 [D] ±1.6	1188.3 [D] ±1.6	0.013030 [N] ±0.000027	1.89 [N] ±0.06	-6.3872 [C]	247.92065 [B]	-1.0 [B]	0.3 [B]	0.62 [*]	1.21 [*]

References

[*] *Value and uncertainty derived from other referenced values and uncertainties in this table*. Bulk Density computed based on the volume of a sphere with the published mean radius.

[B] *Explanatory Supplement to the Astronomical Almanac*. 1992. K. P. Seidelmann, Ed., p.706 (Table 15.8) and p.316 (Table 5.8.1), University Science Books, Mill Valley, California.

[C] Seidelmann, P.K. et al. 2007. "Report of the IAU/IAG Working Group on cartographic coordinates and rotational elements: 2006" *Celestial Mech. Dyn. Astr.* **98**:155-180.

[D] Archinal, B.A. et al. 2018. "Report of the IAU/IAG Working Group on cartographic coordinates and rotational elements: 2015" *Celestial Mech. Dyn. Astr.* **130**:22.

[E] Hilton, J.L. 2005. "Improving the Visual Magnitudes of the Planets in *The Astronomical Almanac*. I. Mercury and Venus" *Astron. J* **129**:2902-2906.

[F] Anderson, J.D., et al. 1987. "The mass, gravity field, and ephemeris of Mercury" *Icarus* **71**:337-349.

[G] Konopliv, A.S., et al. 1999. "Venus gravity: 180th degree and order model" *Icarus* **139**:3-18.

[H] Folkner, W.M. and Williams, J.G. 2008. "Mass parameters and uncertainties in planetary ephemeris DE421." Interoffice Memo. 343R-08-004 (internal document), Jet Propulsion Laboratory, Pasadena, CA.

[I] Jacobson, R.A. 2008. "Ephemerides of the Martian Satellites - MAR080" Interoffice Memo. 343R-08-006 (internal document), Jet Propulsion Laboratory, Pasadena, CA.

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[L] Jacobson, R.A. 2014. "The Orbits of the Uranian Satellites and Rings, the Gravity Field of the Uranian System, and the Orientation of the Pole of Uranus" *AJ* **148**:76-88.

[M] Jacobson, R.A. 2009. "The orbits of the Neptunian satellites and the orientation of the pole of Neptune" *AJ* **137**:4322.

[N] Brozovic, M. et al. 2015. "The orbits and masses of satellites of Pluto" *Icarus* **246**:317-329.

Data updated 2001-May-22:
Sidereal orbit period data were corrected. Previous values were taken from p.704 in reference [B] where they were incorrectly labeled as Sidereal instead of Tropical. The corrected values are derived from the mean longitude rates shown in Table 5.8.1 in reference [B].

Data updated 2002-Nov-22:
Pluto's mass and density were updated using a more current reference. It should be noted that Pluto's radius ranges from about 1150 to 1206 km in the literature.

Data updated 2006-Mar-14:
A previous reference ("Astrometric and Geodetic Properties of Earth and the Solar System" in Global Earth Physics, A Handbook of Physical Constants, AGU Reference Shelf 1, 1995, American Geophysical Union, Tables 6,7,10.) had a few erroneous values. Those values have been replaced using data from the current reference list.

Data updated 2008-Oct-24:
Sidereal rotation periods were updated using improved values from the "Report of the IAU/IAG Working Group on cartographic coordinates and rotational elements: 2006" published in 2007. Rotation rates were significantly improved for Saturn, Uranus, and Neptune. Visual magnitude parameter V(1,0) values were improved for Mercury and Venus using reference [E] listed above.

Data updated 2008-Nov-05:
Mass values were updated from current estimates of GM (referenced above). The value of G (Newtonian gravitational constant) was taken from the current best estimate (CODATA 2006) available from the NIST website, G=6.67428 (±0.00067) x 10^{-11} kg^{-1} m^3 s^{-2}. Mean radius values were added. Bulk density values (and uncertainties) were computed based on updated mass values and computed volumes (based on a sphere of the published mean radius). Equatorial surface gravity values were computed from updated mass and equatorial radius values. Escape velocities were also updated based on computed values using updated masses and mean radii.

Data updated 2018-Mar-28:
Mass values were updated from current estimates of GM (referenced above). The value of G (Newtonian gravitational constant) was taken from the current best estimate (CODATA 2014) available from the NIST website, G=6.67408 (±0.00031) x 10^{-11} kg^{-1} m^3 s^{-2}. Bulk density values (and uncertainties) were computed based on updated mass values and computed volumes (based on a sphere of the published mean radius). Equatorial surface gravity values were computed from updated mass and equatorial radius values. Escape velocities were also updated based on computed values using updated masses and mean radii.